Innate and Adaptive Immunity

Immunology - Lecture 2

Faculty of Medicine

Objectives

- Compare innate and adaptive immunity
- Describe inflammation, phagocytosis and complement
- Describe the roles of T cells, B cells and antibodies

Overview of the Immune System

- Innate immunity is present from birth, responds within minutes to hours and does not improve with repeated exposure.
- Adaptive immunity takes days to develop, is specific to each antigen and forms immunological memory.
- The first line of defence consists of physical and chemical barriers such as intact skin, mucus and stomach acid.

Cells of Innate Immunity

- Neutrophils are the most numerous white blood cells and the first phagocytes to arrive at a site of infection.
- Macrophages develop from monocytes that leave the blood; they phagocytose microbes and present antigens.
- Natural killer (NK) cells kill virus-infected cells and tumour cells without prior sensitization.
- Mast cells release histamine, which increases vascular permeability.

Inflammation

- The cardinal signs of inflammation are redness, heat, swelling, pain and loss of function.
- Histamine and other mediators dilate arterioles and make capillaries more permeable, so fluid and proteins leak into the tissue.
- Neutrophils reach the tissue in steps: margination, rolling along the endothelium, firm adhesion, then migration through the vessel wall (diapedesis).
- Neutrophils then move toward the site of infection by chemotaxis.

Phagocytosis

- The steps of phagocytosis are chemotaxis, adherence to the microbe, ingestion into a phagosome, fusion with a lysosome, and killing and digestion.
- Opsonins such as the complement fragment C3b and IgG antibodies coat microbes and make them easier to ingest.
- Phagocytes kill ingested microbes with reactive oxygen species produced by the respiratory burst.

The Complement System

- The classical pathway is activated by antibodies bound to an antigen.
- The alternative pathway is activated directly on microbial surfaces.
- The lectin pathway is activated when mannose-binding lectin binds sugars on microbes.
- All three pathways meet at the cleavage of C3: C3b acts as an opsonin, C3a and C5a attract inflammatory cells, and C5b to C9 form the membrane attack complex that lyses cells.

Antigen Presentation

- MHC class I molecules are found on all nucleated cells and present peptides from inside the cell to CD8+ T cells.
- MHC class II molecules are found on antigen-presenting cells (dendritic cells, macrophages and B cells) and present peptides to CD4+ T cells.
- Dendritic cells are the most effective antigen-presenting cells for activating naive T cells.

T Lymphocytes

- T cells mature in the thymus.
- Helper T cells (CD4+) release cytokines that activate B cells, macrophages and cytotoxic T cells.
- Cytotoxic T cells (CD8+) kill virus-infected cells by releasing perforin and granzymes.
- Regulatory T cells suppress immune responses and help to prevent autoimmunity.

B Lymphocytes and Antibodies

- B cells mature in the bone marrow; when activated they become plasma cells that secrete antibodies.
- IgM is the first antibody produced in a primary response and exists as a pentamer.
- IgG is the most abundant antibody in serum and the only one that crosses the placenta.
- IgA is found in secretions such as saliva, tears and breast milk; IgE binds mast cells and mediates allergic reactions.

Primary and Secondary Responses

- In the primary response antibodies appear after a lag of about 7 to 10 days, and IgM predominates.
- In the secondary response memory cells produce more antibody, faster, and mainly IgG.
- Vaccination produces memory cells without causing the disease.

Active and Passive Immunity

- Active immunity is produced by the person's own immune system after infection or vaccination, and it lasts a long time.
- Passive immunity is the transfer of ready-made antibodies, such as maternal IgG across the placenta or an injection of antiserum.
- Passive immunity acts immediately but lasts only weeks to months, because no memory cells are formed.

Hypersensitivity Reactions

- Type I (immediate) hypersensitivity is mediated by IgE on mast cells; an example is anaphylaxis.
- Type II is caused by antibodies against cell-surface antigens; an example is a transfusion reaction.
- Type III is caused by immune complexes deposited in tissues; an example is serum sickness.
- Type IV (delayed) hypersensitivity is mediated by T cells and appears after 48 to 72 hours; an example is the tuberculin skin test.

Check Your Understanding

1. Which antibody would you measure to detect a recent infection, and why?
2. Why does a baby have protection against some infections in its first months?

Thank You